

Deepak Mallampati

APPLIED AI ENGINEER

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US-based, F-1 OPT - open to Madrid relocation

PROFESSIONAL SUMMARY

Applied AI engineer with 4+ years building agentic and RAG systems end-to-end for regulated enterprises. Architected conductor-subagent LLM orchestration that cut token consumption 42% at sustained accuracy, and partnered with 20+ internal teams at Vanguard to ship 12 APIs and microservices serving 100K+ requests daily (p95 1.5s to 800ms). Strong in LangChain, multi-agent orchestration, RAG, prompt engineering for structured output, and LLM evaluation (RAGAS, BERTScore), with deployment experience across AWS Bedrock, Azure AI, and GCP Vertex. Comfortable translating business context into working AI prototypes - from healthcare claims processing to enterprise model selection.

CORE COMPETENCIES

Agentic AI / Multi-Agent Orchestration • RAG + Retrieval Ranking • LangChain / LangGraph • Prompt Engineering (Structured Output) • LLM Evaluation + Guardrails • Python / FastAPI • AWS Bedrock / Azure AI / GCP Vertex • Cross-Functional Stakeholder Delivery • Rapid Prototyping

SKILLS

Generative AI & LLMs: GPT-4o, GPT-5, Claude Sonnet, Gemini, LLaMA; LangChain/LangGraph; RAG; prompt engineering; multi-agent orchestration; function calling; fine-tuning (LoRA, QLoRA, PEFT); content safety; Hugging Face Transformers

ML / NLP: PyTorch, TensorFlow, scikit-learn, XGBoost; NER, text classification, summarization; A/B testing, drift detection; RAGAS, BERTScore

Vector Search & Retrieval: FAISS, Pinecone, Weaviate, pgvector; hybrid search; cross-encoder re-ranking; semantic caching

Cloud + MLOps: AWS Bedrock, SageMaker, Lambda, S3; Azure OpenAI, AI Services, Content Safety; GCP Vertex AI; Docker, Kubernetes; Terraform; CI/CD (Jenkins, Azure DevOps); MLflow, W&B

Data: PostgreSQL, MongoDB, Redis; Spark, Databricks; Kafka; pandas, NumPy, pydantic; data lineage and governance

Security & Compliance: HIPAA, SOC 2, PCI DSS; OAuth 2.0, JWT, RBAC; PII redaction; prompt injection mitigation; audit logging

Languages: Python, JavaScript/TypeScript, Java, SQL; FastAPI, React

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present

AI Engineer

USA - Healthcare AI

- Architected production **agentic LLM systems** on a conductor-subagent orchestration framework, decomposing complex objectives into context-scoped subagents - reduced token consumption **42%** while sustaining task accuracy across multi-step workflows.
- Engineered high-performance **RAG pipelines** over large-scale healthcare document corpora, combining dense vector retrieval with cross-encoder re-ranking to lift answer relevance and reduce hallucination on domain-specific queries.
- Designed **privacy-preserving data workflows** for sensitive clinical datasets (K-anonymity, L-diversity, Laplace differential privacy) enabling compliant analytics with zero PII exposure.
- Built fault-tolerant **automation infrastructure** with batch orchestration and exponential-backoff retry; reduced failed production runs **60%**.
- Established **LLM evaluation and monitoring framework** (RAGAS, BERTScore, custom domain metrics) with latency and accuracy dashboards to benchmark iterations and surface regressions pre-release.
- Optimized inference cost and latency via prompt compression, semantic caching, and model routing.

Vanguard

Jan 2024 - Jan 2025

AI Engineer Intern

USA - Financial Services AI Platform

- Developed **AI agents and backend services** for Vanguard's internal AI platform, integrating with production data pipelines and observability tooling across **25+ AI agents** and workflows.
- Optimized prompts and evaluated **GPT-4o, Claude Sonnet, and Gemini**, lifting task success **27%** on internal eval sets and informing platform model-selection decisions.
- Engineered **safeguards and content controls** for policy-compliant AI responses, reducing unsafe outputs **42%** with quality held.
- Built **12 APIs and microservices** powering new AI capabilities, reducing p95 latency from 1.5s to **800ms** and accelerating feature integration **40%**.
- Partnered with product, platform, and data engineering across **20+ internal teams** to deliver scalable LLM applications supporting **100,000+ requests daily**.

Kent State University

Oct 2023 - Jan 2024

ML & Gen AI Research Assistant

USA - Clinical ML + Long-Form Generation

- Built **privacy-preserving ML pipeline** on hospital-readmission data; k-anonymity (k=3), l-diversity (l=2), Laplace DP across 4 epsilon levels.
- Reduced re-identification risk **3.38% to 0.32%** while retaining **99% of baseline predictive performance** (Random Forest AUC 0.689 vs 0.696).
- Fine-tuned **Qwen3-27B via 4-bit QLoRA** on H100 with a 361-example instruction set, conditioning on a six-stage narrative schema for controllable long-form generation.
- Authored IEEE-format conference paper documenting methodology, experiments, and results.

Emerson (Energy Solutions International)

Jun 2022 - Apr 2023

ML Trainee

India - Oil & Gas ERP

- Developed supervised regression and classification models on operational pipeline data for failure prediction, anomaly detection, and capacity forecasting.
- Fine-tuned **BERT and RoBERTa** for entity extraction and text classification on domain-specific financial data; **89% F1** on custom NER.

PROJECTS

career-ops - Autonomous AI Job-Search Pipeline

End-to-end Proof-of-Value system: scans listings, scores fit, generates tailored CVs and applications overnight without supervision. Demonstrates scope - prototype - ship - measure discipline on an agentic AI product.

Node.js, Playwright, multi-agent orchestration, LLM evals

MangaLens - Shipped Chrome Extension

Live on Chrome Web Store, real-time AI translation across **29+ sites**. Multi-provider abstraction (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama) with per-provider payload handling for failure isolation.

TypeScript, Manifest V3, OpenAI / Anthropic / Ollama

Privacy-Preserving Clinical ML Pipeline

Combines k-anonymity, l-diversity, and differential privacy to let researchers analyze sensitive patient data without exposing identities; quantifies privacy-utility tradeoff across six anonymization configurations.

Python, scikit-learn, differential privacy

Story Director LLM

Fine-tuned LLM that generates structured short-form video narratives from a single prompt, paired with an automated rendering pipeline.

Qwen3-27B, QLoRA, H100

EDUCATION

M.S. Computer Science - *Kent State University*

2025

GPA 3.70 / 4.0. Focus: Applied Machine Learning, NLP, Database Systems.